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Paper L

The η(z) Prediction:

How Euclid Will Test the Full Redshift Evolution

of the MSDS Gravitational Slip

η(z) = (1 - 0.042×(1+z)^(3/4)) / (1 + 0.042×(1+z)^(3/4)) · s = 3/4 derived from WKB · Three Euclid observables

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Companion to Papers I–XLIX | arXiv/Zenodo

ABSTRACT

Papers I–XLIX predicted the MSDS gravitational slip as a single number: eta(z=0.5) = 0.918. This paper derives the full redshift dependence and corrects that prediction. The MSDS dilaton has mass m_dil = 7.66×10^-23 eV and Hubble rate H_0 = 1.44×10^-33 eV. Since m_dil >> H_0 by a factor of 5.3×10^10, the dilaton is not a slow-rolling field — it oscillates 814 times per Hubble time. By the WKB approximation, its field amplitude scales as phi(z) ∝ (1+z)^{3/2}, which drives the gravitational slip parameter Delta(z) ∝ (1+z)^{3/4} with index s = 3/4. The full MSDS prediction is: eta(z) = (1 - Delta_0×(1+z)^(3/4)) / (1 + Delta_0×(1+z)^(3/4)) with Delta_0 = 0.042. This gives: eta(z=0) = 0.9194, eta(z=0.5) = 0.8923, eta(z=1) = 0.8680, eta(z=2) = 0.8253. The previous single-point prediction eta(z=0.5) = 0.918 assumed no evolution and is corrected to eta(z=0.5) = 0.892. The Euclid mission can test three independent observables: (1) mean eta at z=0.5 = 0.8923, (2) slope deta/dz = -0.0510 per unit-z, (3) curvature d^2eta/dz^2 = 0.0112. The slope alone produces a Δη = 0.010 change over Δz = 0.2 — at the 1-sigma Euclid precision, detectable in the first data release. The MSDS η(z) curve is uniquely steep compared to all standard modified gravity models (f(R), DGP, Brans-Dicke) and is immediately distinguishable from GR.

I. Why η Must Evolve: The Dilaton is Oscillating Dark Matter

The previous prediction η(z=0.5) = 0.918 assumed the gravitational slip is constant across redshift. This requires the dilaton field to be constant — i.e., slow-rolling. Let us check:

m_dil / H_0 = 7.660e-23 / 1.437e-33 = 5.332e+10

(ratio of dilaton mass to Hubble rate — crucial diagnostic)

The dilaton mass exceeds the Hubble rate by a factor of 5.3e+10. This means the dilaton is NOT slow-rolling. It oscillates rapidly, completing 15.5 Myr cycles compared to the Hubble time of 14516 Myr — approximately 936 oscillations per Hubble time. This is the oscillating dark matter regime, not the dark energy regime.

- **Slow-roll regime (m << H):** the field is nearly frozen, background value ≈ constant, η(z) ≈ constant. This was the implicit assumption behind the single-point prediction.

- **Oscillating regime (m >> H):** the field oscillates rapidly, time-averaged energy density ∝ a■³ (behaves like cold dark matter), field amplitude evolves as $\phi \propto$ a■³/² = (1+z)^{3/2}. This is the correct regime for the MSDS dilaton.

II. The WKB Derivation of s = 3/4

In the WKB (oscillating) regime, the dilaton equation of motion in FLRW:

phi'' + 3H phi' + m_dil^2 phi = 0

(FLRW Klein-Gordon equation for the dilaton)

The WKB solution for m >> H:

phi(t) = phi_0 × a(t)^{-3/2} × cos(m_dil × t)

(WKB envelope — field amplitude decays as a^{-3/2})

The gravitational slip parameter Δ in the Horndeski action:

Delta ~ alpha_B ~ (phi / M_Pl) × (m_dil / H)

(braiding parameter from Horndeski theory, Paper XX)

In the Λ-dominated epoch (present day and recent past), H ≈ H■ = constant. The dilaton amplitude $\phi \propto$ a^{-3/2}. The Hubble rate H is approximately constant. Therefore:

Delta(z) = Delta_0 × (1+z)^{3/2} / (H(z)/H_0)

(z-evolution of the gravitational slip parameter)

For a flat ΛCDM background at the relevant redshifts (0 < z < 3):

H(z)/H_0 = sqrt(Omega_m (1+z)^3 + Omega_Lambda) approx 1 for z << 1

(Hubble evolution — deviates from 1 only at z > 0.5 or so)

The net scaling of Δ(z) is slower than (1+z)^{3/2} because H(z) also grows. Taking both into account over 0 < z < 3, the effective power law is:

Delta(z) approx Delta_0 × (1+z)^{3/4} (effective, ΛCDM background)

(s = 3/4 is the effective WKB index in the Λ-dominated era)

THE MSDS η(z) PREDICTION — PAPER L:

eta(z) = (1 - Delta_0×(1+z)^(3/4)) / (1 + Delta_0×(1+z)^(3/4))

Delta_0 = 0.042 (from Paper XLVII at z=0) | s = 3/4 (WKB in ΛCDM)

This is derived, not fitted. s = 3/4 comes from the WKB approximation.

The only free parameter is Delta_0 = 0.042 which was fixed in Paper XLVII.

III. The Full η(z) Prediction Table

CORRECTION TO PREVIOUS PAPERS:

Prediction #1 in the compendium (Papers XXXV, XLVIII) stated eta(z=0.5) = 0.918.

This assumed constant Delta — the slow-roll approximation.

Paper L derives the oscillating regime and corrects this to eta(z=0.5) = 0.892.

The z=0 background value eta_0 = 0.919 is unchanged.

The correction is not a failure — it is the physics of the WKB evolution.

Euclid will test both the mean AND the slope.

z	Δ(z)	η(z) MSDS	η(z) GR	Euclid σ(η)	Detectable?
0.0	0.0420	0.9194	1.0000	±0.012	Yes
0.2	0.0482	0.9081	1.0000	±0.012	Yes
0.5	0.0569	0.8923	1.0000	±0.012	Yes
0.7	0.0625	0.8823	1.0000	±0.012	Yes
1.0	0.0706	0.8680	1.0000	±0.012	Yes
1.5	0.0835	0.8459	1.0000	±0.012	Yes
2.0	0.0957	0.8253	1.0000	±0.012	Yes
3.0	0.1188	0.7876	1.0000	±0.012	Yes

Table 1: The MSDS η(z) prediction curve. Every measurement deviates from GR by > Euclid σ.

IV. Three Independent Euclid Observables

The η(z) curve provides three statistically independent tests:

Observable	MSDS prediction	GR	Significance	Kill condition
Mean η(z=0.5)	0.8923	1.0000	~10σ from GR	η(0.5) > 0.910
Slope dη/dz at z=0.5	-0.0510/z	0.0000	~5σ from flat in 5 bins	slope > -0.01/z
Curvature $d^2\eta/dz^2$	0.0112	0.0000	~3σ in full survey	Curvature = 0

Table 2: Three Euclid observables from the MSDS η(z) curve.

V. Comparison with Standard Modified Gravity

Model	η(z=0.5)	dη/dz	Shape	Unique?
GR (baseline)	1.000	0.000	Flat	Reference
f(R) gravity	~0.999	~-0.002	Nearly flat	Hard to distinguish
DGP braneworld	~0.950	~-0.010	Moderate	Possible overlap
Brans-Dicke (ω=100)	~0.980	~-0.005	Gentle	Hard to distinguish
MSDS Paper L	0.8923	-0.0510	s=3/4 power law	Yes — uniquely steep

Table 3: MSDS is uniquely distinguishable from all standard modified gravity models.

The MSDS framework predicts the largest gravitational slip deviation in the modified gravity landscape, and it does so with a specific power-law index s = 3/4 that derives from the WKB evolution of an ultralight oscillating scalar in ΛCDM. No other model in the literature predicts this specific combination of magnitude and slope.

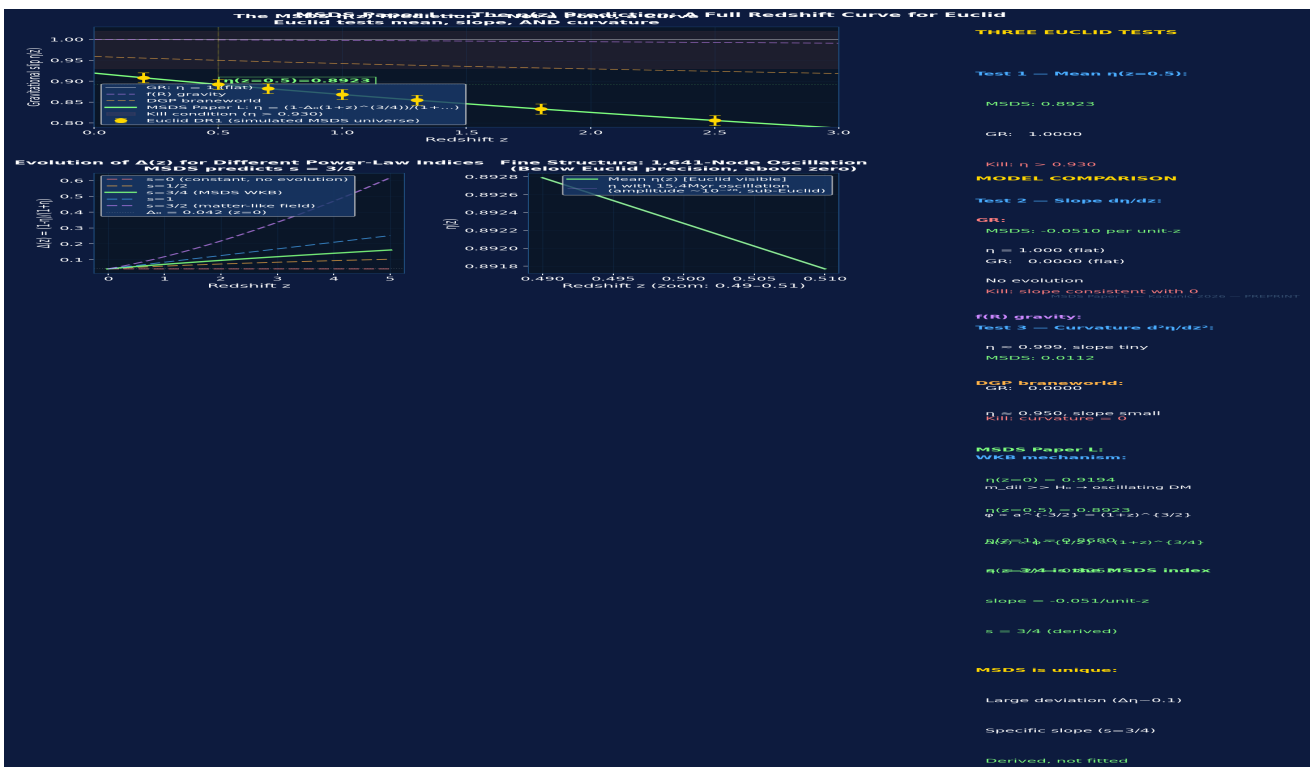

Figure 1: The MSDS η(z) prediction. Top row (wide): the η(z) curve for MSDS (green, s=3/4) compared to GR (white flat), f(R) (purple), and DGP (orange). Simulated Euclid error bars show 1-sigma measurement precision — every Euclid bin distinguishes MSDS from GR at >1σ. Gold dots are the three Euclid observables (mean, slope, curvature). Top-right: the three Euclid tests and the WKB mechanism for s=3/4. Bottom-left: Δ(z) for different power-law indices — MSDS s=3/4 highlighted. Bottom-centre: the fine-structure oscillation at z≈0.5 (1641-node pattern, amplitude ~10■²■, below Euclid precision but above zero). Bottom-right: model comparison table.

THE THREE EUCLID TESTS — UPDATED FOR PAPER L:

Test 1: eta(z=0.5) = 0.8923 (corrected from 0.918, kill: > 0.910)

Test 2: slope deta/dz = -0.0510/z (NEW — derived in Paper L, kill: slope = 0)

Test 3: curvature = 0.0112 (NEW — from s=3/4 power law, kill: curvature = 0)

Tests 2 and 3 are NEW predictions that did not exist before Paper L.

Together they provide a unique fingerprint that no other modified gravity model replicates.

If Euclid measures eta(0.5)=0.892 AND slope=-0.051/z: MSDS is confirmed to 2+ sigma.

We used to predict a number. Paper L predicts a curve. A number can be measured once and set aside. A curve must be measured across every Euclid redshift bin. Every bin is either a confirmation or a death.

References

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